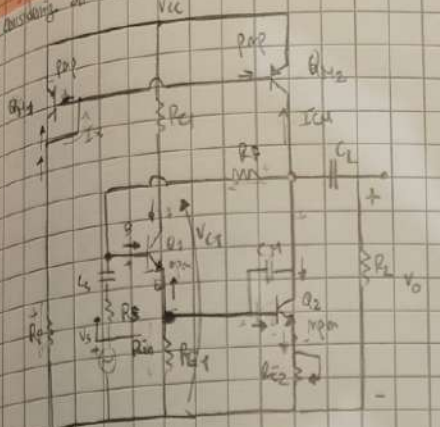


8/07/2022

considering all currents entering the transistor

Ex1



$$I_{C1} = \frac{V_{CC} - V_{CE1}}{R_{C1}} = 2 \text{ mA}$$

$$I_{B1} = \frac{I_{C1}}{\beta_{F1}} = 10 \mu\text{A}$$

$$I_E = -I_{C1} \left(1 + \frac{1}{\beta_{F1}} \right) = -2.01 \text{ mA}$$

$$\frac{V_{CC} + V_{CE1} \text{ pop}}{R_p} = I_{C1}$$

$$I_{B1} = -I_{B2} = I_{C1} - I_{C2}$$

$$I_{C1} + 2I_{C2} = -I_{C1} \rightarrow I_{C2} = -I_{C1}$$

$$I_{C2} = \frac{-(V_{CC} + V_{CE1} \text{ pop})}{R_p} \frac{\beta_{F1}}{2 + \beta_{F1}} = -1.953 \text{ mA}$$

$$V_{CE1} = V_{CC} - R_{C1}(I_{C1} + I_{B1} - I_{B2})$$

$$I_{C2} = -I_{C1} - I_{B1} = 1.943 \text{ mA}$$

$$I_{B2} = \frac{I_{C2}}{\beta_{F2}} = 9.72 \mu\text{A}$$

$$V_{CE2} = V_{BE1} + V_{BE2} + R_{E1}I_{B1}$$

$$V_{CE1} = 6 \text{ V} > V_{CE1 \text{ sat}}$$

$$V_{CE2} = 3.3 \text{ V} > V_{CE2 \text{ sat}}$$

$$R_{E2} = \frac{V_{C1} - V_{CE1} - V_{BE2}}{I_{C1} + I_{B2}} = 691 \Omega$$

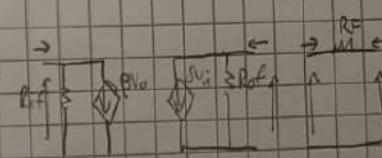
G_s, R_s shorted, G_m opened

Partial topology: current mirror and voltage divider

Not an equivalent and the input mirror effect is a bit, looking

at $r_{\pi} = r_o = \infty \rightarrow$ open circuit, ideal current generator

$$\begin{aligned} A_{if} &= R_F \\ A_{if} &= R_F \\ \beta &= -1 \\ R_F &= R_F \end{aligned}$$

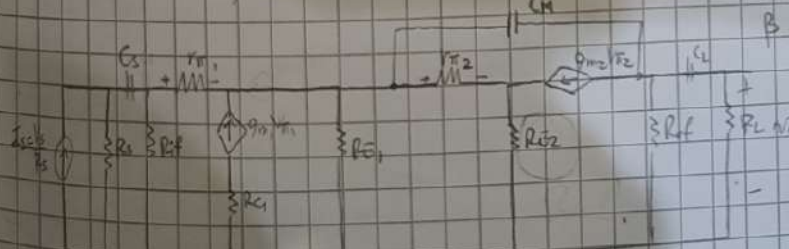


$$g_{m1} = \frac{I_{C1}}{V_T} = 0.0769 \text{ S}$$

$$R_1 = 2.6 \text{ k}\Omega = \frac{R_{C1}}{g_{m1}}$$

$$g_{m2} = \frac{I_{C2}}{V_T} = 0.0717 \text{ S}$$

$$R_2 = \frac{R_{C2}}{g_{m2}} = 2.68 \text{ k}\Omega$$



$$A_{VCE1} = \frac{-\beta_{F1}(R_{C1} \parallel R_L)}{r_{\pi1} + (1 + \beta_{F1})R_{E1}} = -23.3$$

$$A_{VCE2} = \alpha_{F2} \alpha_{F1} A_{VCE1} = -209.3 \text{ k}\Omega$$

$$R_{inCE2} = r_{\pi2} + (1 + \beta_{F2})R_{E2} = 141.6 \text{ k}\Omega$$

$$A_F = \frac{A_{VCE2}}{1 + \beta_{F2}} = -102.3 \text{ k}\Omega$$

$$\alpha_{F1} = (1 + \beta_{F1}) \frac{R_{E1} \parallel R_{inCE2}}{r_{\pi1} + (1 + \beta_{F1}) \frac{R_{E1} \parallel R_{inCE2}}{1 + \beta_{F1}}}$$

$$A_V = \frac{V_o}{V_s} = A_F \frac{1}{R_s} = -10.23$$

$$R_{inCE1} = r_{\pi1} + (1 + \beta_{F1})R_{E1} \parallel R_{inCE2} = 202.6 \text{ k}\Omega$$

$$R_{in} = R_s \parallel R_{inCE1} \parallel R_{inCE2} = 3.1 \text{ k}\Omega$$

$$R_{in}^a = R_s \parallel R_{inCE1} \parallel R_{inCE2} = 3.1 \text{ k}\Omega$$

$$R_{in}^b = \frac{R_{in}^a}{1 + \beta_{F1}} = 1.45 \text{ k}\Omega$$

$$R_{in} = \frac{1}{\frac{1}{R_{in}^b} + \frac{1}{R_s}} = 13.02 \text{ k}\Omega$$

considering only the effect of $C_M \rightarrow f_H = ?$

OTC method

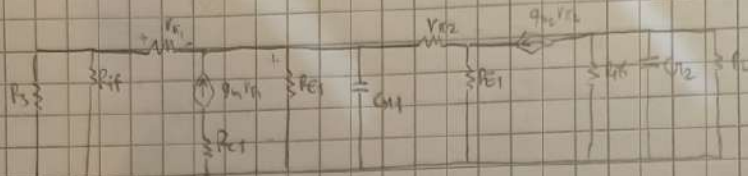
$$f_H = \frac{1}{2\pi C_M R_{M,OC}}$$

we can separate the C_M cap using Miller theorem

$$C_{M1} = C_M (1 - G)$$

$$C_{M2} = C_M (1 - \frac{1}{G})$$

$$G = \frac{C_M}{C_{M,open}} = d.c.g. = -23.3 \checkmark$$



when working with A_1 in this case we will not determine what we expect because the caps are in a place where the voltages seen from the caps are reduced so f_H is increased $\rightarrow$ (opposite of $f_H = \frac{1}{2\pi R_{M,OC} C_M}$)

$$f_H = \frac{1}{2\pi C_M R_{M,OC} + C_{M2} R_{M,OC}}$$

$$R_{M,OC} = R_{i1} \parallel R_L = 16.51 \text{ k}\Omega$$

$$R_{M,OC} = R_{i1,OC} \parallel R_{o1,OC}$$

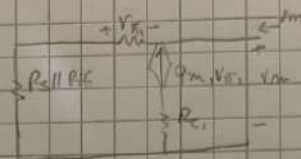
$$\downarrow 6028 \Omega \checkmark$$

$$C_{M1} = 1.215 \text{ nF} \checkmark$$

$$C_{M2} = 52.1 \text{ pF} \checkmark$$

$$f_H = 120.5 \text{ kHz} \checkmark$$

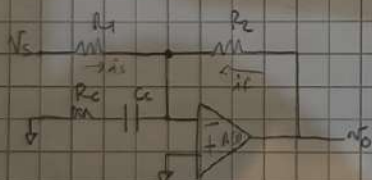
$$f_H^F = (1 + B A_{CL}) f_H = 369 \text{ kHz} \checkmark$$



can't use the feedback to look for f_H

$$v_{in} = \frac{-v_o - \frac{v_o}{R_L} R_s \parallel R_i}{\frac{v_o}{R_L} - g_m v_o} = \frac{v_o + R_s \parallel R_i}{(1 + \beta a_1)} = 60.3 \text{ mV} \checkmark$$

EX2

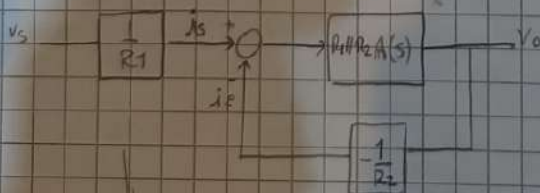


$$A(s) = \frac{A_{d0}}{(1 + \frac{s}{\omega_1})(1 + \frac{s}{\omega_2})}$$

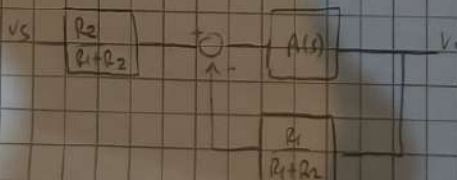
non ideal, partially compensated opamp

with C_c ignored, $A_v(s) = \frac{v_o}{v_s} = -\frac{R_2}{R_1}$ inverting configuration of A_v opamp

Feedback theory: $R_2 \rightarrow$ feedback network voltage sampling - current mixing



direct loop block diagram



$$\beta = \frac{R_1}{R_1 + R_2} = 0.318$$

A.18)

$$V_{CC} = 24 \text{ V}$$

$$V_{BE} = +0.7 \text{ V}$$

$$\beta_F = 200$$

$$1) V_B = V_{CC}$$

$$V_B = R_{B1} \cdot V_{CC}$$

$$V_B = V_{BE} + I_B R_{B2}$$

$$\frac{R_{B1} + R_{B2}}{\beta_F} \cdot I_B$$

$$V_{BE} = V_{CC}$$

$$2) V_S = V_{CC}$$

frequency bandwidth and PM

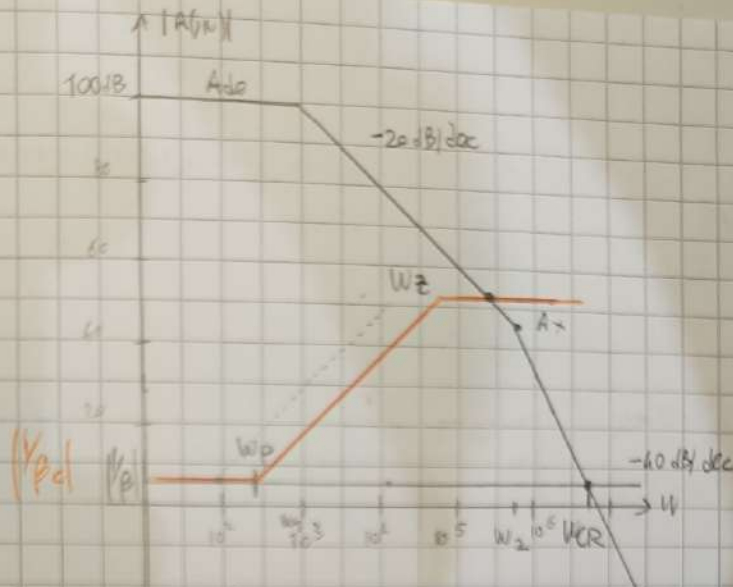
$$\frac{1}{\beta} = 5 \text{ dB}$$

$$A_{OL} W_1 = A \times W_2 \rightarrow A \times = \frac{A_{OL} W_1}{W_2}$$

$$A \times W_2^2 = \frac{1}{\beta} W_{CR}^2$$

$$\sqrt{\beta A_{OL} W_1 W_2} = W_{CR} = 3.99 \text{ Mrad/s}$$

$$W_{CR} \approx 4 \cdot 10^6 \text{ rad/s} = \text{Bandwidth}$$



$$PM = 180^\circ - \arctan\left(\frac{W_{CR}}{W_2}\right) - \arctan\left(\frac{W_{CR}}{W_1}\right) = 7.16^\circ$$

Applying the $R_C - C_C$ compensation network pole and zero

When $W \rightarrow \infty$ we add a R_C in parallel to R_1 so $\beta_{DC} = \frac{R_C / R_1}{R_C / R_1 + R_2} = 3.87 \cdot 10^{-3}$

$$\frac{1}{\beta} \cdot \frac{1}{W_p} = \frac{1}{\beta_{DC}} \cdot \frac{1}{W_z}$$

$$W_p \rightarrow \text{LF pole} \rightarrow W_p = \frac{1}{C_C R_{eq}} = \frac{1}{C_C (R_C + R_1 / R_2)} = 259 \text{ rad/s}$$

$$W_z = \frac{\beta}{\beta_{DC}} W_p = \frac{\beta}{\beta_{DC}} \cdot \frac{1}{C_C (R_C + R_1 / R_2)} = 21.259 \text{ Krad/s}$$

$$W_{CR}: A_{OL} W_1 = \frac{1}{\beta_{DC}} W_{CR} \rightarrow W_{CR} = \beta_{DC} A_{OL} W_1 = 387 \text{ Krad/s}$$

$$\text{Bandwidth} = W_{CR} = 387 \text{ Krad/s}$$

$$PM = 180^\circ - \arctan\left(\frac{W_{CR}}{W_1}\right) - \arctan\left(\frac{W_{CR}}{W_2}\right) + \arctan\left(\frac{W_{CR}}{W_z}\right) - \arctan\left(\frac{W_{CR}}{W_p}\right) = 49.5^\circ$$

because we should consider the p not $\frac{1}{\beta}$

A.18)

$$V_{CC} = 24 \text{ V}$$

$$V_{BE} = 0.4 \text{ V}$$

$$\beta = 200$$

$$R_1 = 390 \text{ K}\Omega$$

$$R_2 = 68 \text{ K}\Omega$$

$$R_3 = 56 \text{ K}\Omega$$

$$R_E = 18 \text{ K}\Omega$$

$$R_C = 10 \text{ K}\Omega$$

$$R_4 = 10 \text{ K}\Omega$$

$$C_1 = 10 \mu\text{F}$$

$$C_2 = 22 \mu\text{F}$$

$$C_3 = 4.7 \mu\text{F}$$

$$V_E = V_{CC} \cdot \frac{R_2}{R_1 + R_2} = 3.56 \text{ V}$$

$$R_{TH} = R_1 || R_2 = 52.9 \text{ K}\Omega$$

1 ora e 40 minuti!

